



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 10 • STANDARD MPP.7

Math is Beauty

Lesson 10-2 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can use **bilateral symmetry** to describe structure in nature and in human creations.

Language Objective: I can describe a symmetric design using the words **mirror image**, **line of symmetry**, and **balance**.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

symmetric

mirror image

bilateral symmetry

line of symmetry

balance



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

Having two halves that match as mirror images is being

2

A copy of a figure flipped across a line is its

3

Reflective symmetry in plants and animals is called

4

The line you draw so each side reflects the other is the

5

An even arrangement where neither side feels heavier is

.

Write About the Math

How do you respond to your friend?

¿Cómo le respondes a tu amigo?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **symmetric**
simétrico

☐ **mirror image**
imagen reflejada

☐ **bilateral symmetry**
simetría bilateral

☐ **line of symmetry**
eje de simetría

☐ **balance**
equilibrio

Model: *A figure is symmetric only if there is a specific line where one side is a mirror image of the other. — A strong answer explains the line gives you something to compare the two sides across — without it, you can't test whether the sides actually match. A weak answer just says 'it looks the same.'*

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **To settle the disagreement, draw a ____ through the center of each flower and check whether one side is a ____ of the other. If the test works for both, then ____ flowers have bilateral symmetry.**
- **My answer is ____ because ____.**

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **My evidence is ____, so ____.**
Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** symmetric
2. **Notes 2:** mirror image
3. **Notes 3:** bilateral symmetry
4. **Notes 4:** line of symmetry
5. **Notes 5:** balance
6. **Watch for this mistake:** *Thinking a figure lacks bilateral symmetry just because its line of symmetry is not vertical — the mirror line can run in any direction, and the test is whether the two sides match across it.*
7. **Try It 1:** C. 46 flowers — *One side uses $14 + 9 = 23$ flowers, and the mirror side uses the same 23, so the whole design uses $23 \times 2 = 46$ flowers.*
8. **Try It 2:**